



Analyzing wildfire evacuation dynamics with agent-based modeling in damaged road networks

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ABSTRACT

Wildfires increasingly threaten residents in the Western United States. Despite numerous state and local initiatives aimed at mitigating these risks, completely eliminating the wildfire dangers remains unfeasible due to substantial inherent uncertainties. In this case, evacuation is the most important and effective strategy for reducing human casualties during wildfire events. While the primary goal of evacuation—moving people at risk to safer places—appears straightforward to achieve, the reality is complicated by unpredictable human behaviors and the surge in travel demand, which often results in severe traffic congestion and, consequently, a heightened risk to human lives. In addition, the reduced traffic-carrying capacities of road segments due to wildfires further exacerbate these challenges. In this context, wildfire evacuation simulation can serve as an effective experimental means for emergency management and evacuation planning, offering a cost-effective method to identify bottlenecks and critical congestion points during an evacuation.

This paper proposes an agent-based modeling (ABM) framework specifically designed to simulate wildfire evacuations in damaged transportation settings. The proposed framework uniquely integrates wildfire simulation and road network vulnerability assessment with ABM, allowing for a detailed representation of human behaviors during evacuations and the dynamic network functionality in microscopic traffic simulation. A notable contribution of this study is its fully probabilistic approach, which evaluates evacuation performance and identifies critical components of the road network not under a single scenario but under a range of representative scenarios. This probabilistic perspective provides a more comprehensive understanding of potentially vulnerable and congested points, thereby enabling emergency managers and transportation planners to better allocate resources and enhance mobility during wildfire evacuations. The effectiveness of the ABM framework is demonstrated through its application in simulating wildfire evacuations in the City of Santa Clarita, California. The simulation results aid in both pre-fire planning and emergency decision-making, ultimately contributing to improved evacuation strategies and public safety during wildfire events.

1. Introduction

Recent wildfires in the Western United States have demonstrated their destructive effects on property, human lives, and the environment. The 2018 Camp Fire, the deadliest and most destructive wildfire in California's history, claimed 85 lives and destroyed over 18,000 buildings (Syphard et al., 2022). In 2020 alone, more than 100 wildfires in the Western United States scorched approximately 8.2 million acres of land (Keeley & Syphard, 2021). Moreover, the United States Forest Service (USFS) warns that billions of acres across the country are at risk of catastrophic wildfires. The increasing frequency and severity of

wildfires in the United States are partly attributed to global climate change, which contributes to more frequent heatwaves and prolonged dry periods (Westerling et al., 2006). In addition to climate change, rapid population growth and residential development at the wildland-urban interface (WUI) have greatly intensified property destruction and the human and economic toll of wildfires (Lee et al., 2022; Ma et al., 2022).

Having seen the enormous economic and human losses that wildfires have caused over the past decades, the federal and local governments, researchers, engineers, and the public have paid increasing attention to wildfire risk management. As part of this effort, a variety of community-

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level wildfire protection plans have been proposed and implemented by federal, state, and local government agencies. The plans include zoning regulation, fire department support, adequate water supply and flow for firefighting, road improvement for facilitated access, and public education and outreach (Communities Committee, 2004). In WUI areas, individual homeowners adopt wildfire risk mitigation actions including the use of nonflammable or ignition-resistant materials, the installation of dual-pane windows or fire sprinklers, vegetation fuel treatment, etc. While wildfire risks can be mitigated through these risk mitigation actions, residual risks always exist due to substantial uncertainties in wildfire activity. In this case, evacuation is the most important and effective method to reduce human losses during a wildfire event. For example, while most tsunami risk reduction measures were destroyed during the 2011 Great East Japan Earthquake/Tsunami, rapid evacuation saved approximately 90 % of the total population at risk from the tsunami (Mas et al., 2012). However, in the real world, mass evacuation and traffic jams caused by exit route shortages under simultaneous evacuation orders may complicate the simple and straightforward goal of evacuation that moves people at risk to safer places, thereby jeopardizing human lives (Cova et al., 2005). Although traffic jams during the 2018 Camp Fire were not significant contributors to fatalities, they did result in vehicles being abandoned and residents evacuating on foot, highlighting that road systems were not designed for such an urgent evacuation during a hazard event. A similar situation occurred during the 2023 Maui fire, where some victims were tragically discovered in their vehicles while attempting to evacuate (Alvarez & Fuller, 2018; Clayton et al., 2023). This problem can be extended to other communities at great risk of wildfires in the United States. Thus, effective community-based transportation evacuation planning has become an emergent issue in disaster emergency and risk management.

1.1. Evacuation simulation modeling

A considerable body of research has been carried out to develop transportation evacuation simulation models for various types of natural hazard events. Early studies on emergency evacuation employed macroscopic traffic simulations to capture the dynamic flow of traffic across networks during the evacuation process (e.g., Burkard et al., 1993). However, many of these early models assumed homogenous behaviors among evacuees due to a high computational cost, resulting in evacuation time estimates at the macro- or meso- scale. Since the mid-1990 s, advancements in computer technology have enabled the utilization of microscopic traffic simulations in emergency evacuation models so as to simulate traffic flows at the individual vehicle level (Benjaafar et al., 1997; Hamacher & Tjandra, 2001; Wolshon & Marchive III, 2007).

Several studies started to incorporate the behaviors of individual evacuees during disasters into behavior-based microscopic traffic simulation models. During an evacuation, individual behavior plays a critical role in the evacuation process because it significantly affects pre-movement phases, reaction time, decisions to evacuate, and evacuation phases, thus affecting total evacuation time (D'Orazio et al., 2014). Using the parameters describing each evacuee's characteristics (e.g., start location, physical ability, reaction time, transportation mode, route choice mechanism, vehicle speed, etc.), individual and collective behaviors can be captured in dynamic traffic simulations at the micro-scale (Sinuany-Stern & Stern, 1993).

In recent years, agent-based modeling (ABM) has received great attention due to its advantages in capturing individual and collective behaviors in a dynamic complex system (Cimellaro et al., 2017; D'Orazio et al., 2014; Epstein, 1999; Feng et al., 2020; Mas et al., 2012). ABM is one of the heuristic computational models that can simulate the action and interaction of individual agents with the goal of finding system movement trends. In ABM, each evacuee (or vehicle) is modeled as an autonomous agent who has his/her own characteristics and follows specific decision rules. Individual agents make decisions based on

interactions with other agents and the environment as well as localized knowledge (Chen & Zhan, 2014). Thus, ABM can model individual agents' behaviors simultaneously and capture their collective behavior subsequently in a dynamic system. A large number of studies have employed ABM in modeling evacuation. Pan (2006) investigated human behaviors during an earthquake event using basic concepts of psychology and sociology and incorporated them into multi-ABM for the egress analysis of a multi-story university building. Chen and Zhan (2014) utilized ABM to incorporate the effect of individual drivers' behaviors on traffic flow in three different road structures. D'Orazio et al. (2014) modeled the post-earthquake evacuation patterns of building occupants using videotape analysis.

Several researchers combined hazard models with ABM to represent more realistic evacuation situations. Mas et al. (2012) proposed an integrated ABM to estimate tsunami/earthquake evacuation patterns and validated this model using the 2011 Great East Japan earthquake event having a magnitude of 9.0 Mw. The proposed evacuation model was integrated with a numerical simulation of tsunamis to estimate tsunami propagation and the associated start time of evacuation. Cimellaro et al. (2017) utilized ABM to simulate building evacuation during an earthquake event. The model first performed a numerical structural analysis of a building to estimate structural responses to a wide range of possible earthquake ground motion intensities and incorporated human anxiety and behavior under emergency.

1.2. Wildfire evacuation modeling and framework

Advantaged by the development of microscopic traffic simulation models, the majority of wildfire evacuation modeling has been conducted using these microscopic traffic simulations. Wolshon and Marchive (2007) utilized the CORSIM microscopic simulation platform to conduct traffic simulations during wildfires, exploring various evacuation scenarios to assess the effects of development density and street network layout. de Araujo et al. (2014) employed a metropolitan planning organization four-step model to evaluate a worst-case scenario and compared it with the post-event report of an actual wildfire evacuation in the same area. Their study demonstrated the advantages of using traffic simulation to aid evacuation planning. Chen et al. (2020) employed SUMO microscopic traffic simulation during wildfire evacuations. Similar to evacuation modeling for other types of hazards (as described in Section 1.1), community-based wildfire evacuation modeling has also increasingly utilized ABM. For example, Li (2022) proposed a data-driven wildfire evacuation modeling approach using the ABM platform MATSim and demonstrated a strong correlation between their estimated evacuation times and actual outcomes in a case study. This work underscored the potential of utilizing ABM in wildfire evacuation simulations.

While these microscopic traffic simulations and ABM have significantly advanced our understanding of evacuation dynamics, they may not fully capture the complexities of wildfire scenarios. Wildfires are characterized by rapidly changing microenvironmental conditions (e.g., wind speed and direction), which can force residents to evacuate with little or no notice. Thus, weather analysis and wildfire propagation prediction should be incorporated into the evacuation simulation to improve its accuracy. Although most existing studies have not accounted for hazard analysis (or have utilized very elementary hazard modeling) in their evacuation models (Beyki et al., 2023; Wahlqvist et al., 2021), some recent research has begun to address this gap by integrating more sophisticated wildfire simulation model. For example, Wahlqvist et al. (2021) proposed a wildfire evacuation modeling platform WUI-NITY, which was built on the Unity3D game engine. The platform integrated a wildfire modeling layer based on FARSITE, a traffic sub-model, and a simple response model. The inclusion of the fire layer enabled the simulation of a staged evacuation approach, while an empirical probability distribution was utilized in the pedestrian model to represent response time.

In addition to accurate hazard modeling and microscopic traffic simulation, understanding human behaviors is crucial for improving the performance of evacuation assessments. Partly due to the complex and unpredictable nature of human responses during emergencies, wildfire evacuation behavior has often been studied separately from evacuation simulation modeling. Behavioral models for wildfire evacuation typically focus on binary choices, such as the decision to evacuate or stay (Kuligowski et al., 2020; Kuligowski et al., 2022; Mozumder et al., 2008; Toledo et al., 2018; Walpole et al., 2020; Wong et al., 2022). These models are often informed by qualitative (Christianson et al., 2019; Cohn et al., 2006; McGee et al., 2019; Taylor et al., 2005;) or quantitative data (Koksal et al., 2019; Kuligowski et al., 2022; McCaffrey et al., 2018; Mozumder et al., 2008; Roberson et al., 2012; Toledo et al., 2018). However, this approach often overlooks the complexity of real-world evacuation scenarios. In regions like California, Oregon, and Washington, evacuation orders are frequently issued in stages, with two-level or three-level evacuation announcements being common (Kuligowski et al., 2023). This staged approach necessitates a more advanced understanding of evacuation behavior, where decisions go beyond the simple choice to stay or evacuate. The spatiotemporal dynamics of a wildfire, particularly the movement of the firefront, should directly influence the issuance of staged evacuation orders. These orders, in turn, significantly impact the evacuation behaviors of individuals and communities. Thus, in combination with wildfire simulation, it is essential to incorporate a staged evacuation behavior model into evacuation modeling, as it better reflects the decision-making processes individuals and communities face during these multi-phased evacuation events.

While the integration of wildfire simulation, staged evacuation behavior modeling, and microscopic traffic simulation has been recognized as essential for effective wildfire evacuation modeling, there is another critical component that must not be overlooked: road network vulnerability. Beloglazov et al. (2016) took a significant step forward by developing a wildfire evacuation model that combines spatiotemporal firefront with evacuation trigger modeling and ABM. Despite being one of the most comprehensive wildfire evacuation models, it still lacked a vulnerability component. In addition, the data and parameters used in the trigger and behavioral modeling were assumed or determined based on expert opinions, which made them hardly reproducible. To date, most wildfire evacuation simulation models fail to account for changes in network connectivity during an event, despite its critical impact on evacuation performance and evacuee safety (Niu et al., 2022). This oversight is particularly concerning given the strong negative correlation between network connectivity and fatalities. Given that many parts of a transportation system are vulnerable to natural hazards, their physical damage and the associated reduction in traffic carrying capacity due to a hazard event should be considered in a community-based evacuation model so as to better represent traffic conditions and evacuation process during the event (Lee and Ma, 2023). To this end, the conventional ABM should be integrated with hazard analysis, behavioral modeling, and vulnerability assessment of a transportation system while being supported by reproducible, quantitative data.

1.3. Research gaps and contributions

The literature review conducted in this section has highlighted several significant research gaps in current wildfire evacuation modeling. First, many existing wildfire evacuation models have not incorporated advanced hazard modeling, thus lacking the ability to provide hazard inputs. This deficiency reduces the accuracy of the models in predicting the diverse conditions that evacuees may face during an actual wildfire. Second, very few studies have addressed the vulnerability assessment of the transportation network, an aspect that is crucial for evaluating evacuation performance under conditions where the network has been damaged. Third, most studies have assumed a uniform evacuation rate or relied on a generalized probability curve, such as the S-curve, to represent evacuation behavior. This is partly

because wildfire behavior models are often developed separately from wildfire evacuation simulation models. This assumption on a uniform or simple evacuation rate fails to capture the variability in individual responses. Integrating demand generation based on individual characteristics would lead to a more heterogeneous and realistic representation of evacuation behavior, more accurately reflecting the diversity of responses observed in real-world scenarios. Finally, most existing evacuation models have relied on a scenario-based approach, which falls short of providing a comprehensive assessment within a full probabilistic space. This limitation hinders the models' ability to fully capture the variability and uncertainty inherent in real-world wildfire events.

This paper proposes an ABM framework for wildfire evacuation in damaged transportation settings. To fill the research gaps in the literature, this study develops an integrated framework that incorporates hazard modeling, vulnerability assessment, evacuee response modeling, and traffic simulation to predict traffic conditions during an evacuation and identify the critical parts of the transportation network for pre-fire risk mitigation actions. The contribution of this study is threefold: (a) the framework incorporates an advanced wildfire hazard modeling and vulnerability assessment to improve the accuracy of wildfire evacuation in damaged transportation settings; (b) this study constructs an evacuee response model based on a stated preference survey to predict individual evacuees' behavior as a firefront approaches; and (c) this study introduces a novel approach by generating representative wildfire events within the domain of probability space, which enables the probabilistic evaluation of evacuation performance while quantifying uncertainties inherent to hazardous events.

2. Proposed methodology: An integrated wildfire evacuation simulation

This study develops an ABM framework for wildfire evacuation in damaged transportation settings aimed at predicting traffic conditions during an evacuation and identifying the critical parts of the transportation network for pre-fire risk mitigation actions. The framework integrates wildfire simulation and vulnerability assessment with ABM to adequately represent both human behavior during an evacuation and time-dependent network functionality in microscopic traffic simulation. More specifically, as illustrated in Fig. 1, the framework consists of four modules: wildfire simulation, vulnerability assessment, evacuee response model, and traffic simulation. The first module evaluates the spatiotemporal probability of wildfire occurrence and generates representative wildfire scenarios in probability space. FARSITE simulates a time-dependent fire-front movement for each scenario and feeds that information into the subsequent modules. The second module performs a road closure assessment to evaluate wildfire-induced changes in traffic capacity. The third module constructs an evacuee response model based on a stated preference survey to predict individual evacuees' behaviors in response to fire propagation. The fourth module simulates traffic conditions by updating traffic demand and capacity at every time step.

All the variables considered in the framework are updated at every fixed time step: road closure status and its traffic carrying capacity are updated based on fire propagation measured at every time step; each agent updates its state during an evacuation; and microscopic traffic simulation coupled with ABM is performed based on the time-dependent network functionality and the updated locations of all agents. Private vehicles are the primary form of mobility for people living in wildfire-prone areas (Beloglazov et al., 2016). We therefore focus on capturing vehicle use and vehicular traffic in our modeling without considering pedestrian behaviors. The final results are time-dependent traffic maps to identify the critical parts of the transportation network that are the most vulnerable to wildfires and have great potential for causing traffic congestion during an evacuation. Moreover, the total number of evacuating vehicles during a given time period will also be obtained as a final result. These results, derived from a probabilistic approach, can be used to determine the road segments that most constrain evacuation

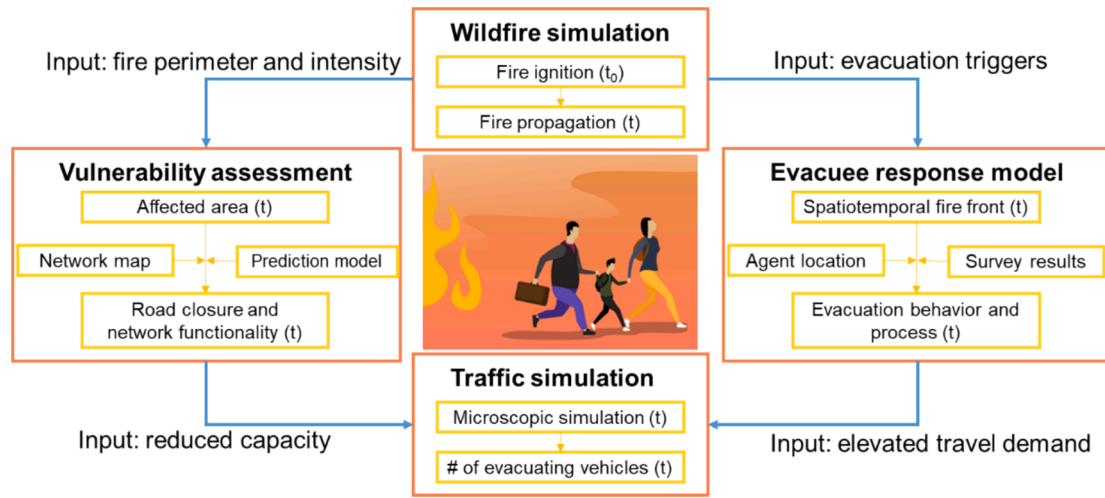


Fig. 1. The proposed ABM framework for wildfire evacuation in damaged transportation settings.

performance and enable targeted interventions to enhance overall evacuation efficiency and safety.

2.1. Module 1: Wildfire simulation

As illustrated in Fig. 1, the first module within this framework pertains to wildfire simulation. This module is responsible for quantifying the inherent uncertainties associated with wildfire occurrences and generating a set of representative wildfire scenarios within the probability space. These scenarios serve as crucial inputs for the subsequent modules, forming the foundational basis for the entirety of the simulations. This module consists of two stages: wildfire occurrence probability estimation and fire growth modeling. The first stage estimates spatiotemporally varying fire occurrence probabilities in the study region and generates representative wildfire scenarios. The second stage simulates wildfire growth for each scenario and records time-dependent firefront that will be fed into the subsequent modules.

In the first stage, a community of interest and its surrounding areas are defined and divided into small grids (e.g., 4 km x 4 km) to find the probabilities of fire that can be propagated into the community. A model for estimating fire occurrence probability should consider two main causes of wildfire, including human causes (e.g., power lines, equipment failure, accidental ignitions, smoking, fireworks, campfires, arson) and natural causes. Population density, urban development, distance to freeway, railway or power lines, etc. can be used to estimate the occurrence rate of human-caused fires (Massada et al., 2012; Rodrigues

single indicator, the energy release component (ERC). The ERC serves as a composite fuel moisture index, representing the variability in both live and dead fuels. The ERC accounts for both short-term and long-term variations in fuel moisture content, which are influenced by changes in precipitation, temperature, and humidity. It has been shown to have a strong correlation with fire occurrence across different climate zones in the United States (Andres et al. 2023; Finney et al., 2011), making it a suitable proxy for representing other weather-related temporal variables.

A generalized additive logistic regression is employed to construct a wildfire occurrence probability estimation model that accounts for spatiotemporal dependence and non-linear relationship between independent and dependent variables (Preisler et al., 2004). Since the wildfire occurrence probability estimation models are used to generate representative wildfire scenarios that trigger mass evacuation, the unconditional probability of a fire occurring and turning into a “large fire” should be estimated. In this study, a large fire is defined as a fire that has burned areas greater than 100 acres (Preisler et al., 2004). The unconditional probability (P_{large}) that a large fire occurs at a given grid cell and a specific ERC value is the product of the ignition probability, P_{ign} , and the conditional probability of an ignition turning into a large fire, $P_{large|ign}$. The generalized additive logistic regression models for estimating these probabilities are expressed mathematically by:

$$P_{ign}(ERC, x, y) = \beta_0 + f_1(ERC) + f_2(x, y) + f_3(pd_{x,y}) + f_4(d_{x,y}^e, d_{x,y}^s) + f_5(cc_{x,y}) + f_6(d_{x,y}^r) \quad (1)$$

$$P_{large|ign}(ERC, x, y) = \gamma_0 + g_1(ERC) + g_2(x, y) + g_3(pd_{x,y}) + g_4(d_{x,y}^e, d_{x,y}^s) + g_5(cc_{x,y}) + g_6(d_{x,y}^r) \quad (2)$$

& De la Riva, 2014), whereas weather and topographic factors (e.g., wind, temperature, relative humidity, landcover, elevation, slope) may affect the occurrence rate of nature-caused wildfires (Finney et al., 2011; Massada et al., 2012). In this study, the distance between the centroid of a grid cell to transmission lines, major roads, picnic/camp sites, the latitude and longitude of the centroid of a grid, the canopy cover ratio, and the population density in a grid were considered as the spatial variables that incorporate both human- and nature-related spatial diversity. On the other hand, temporal variation was encapsulated into a

$$P_{large} = P_{ign} \bullet P_{large|ign} \quad (3)$$

where β_0 and γ_0 = the intercept terms; $f(\bullet)$ and $g(\bullet)$ = the non-parametric smooth functions; ERC = the energy release component; x = the latitude; y = the longitude; $pd_{x,y}$ = the population density at a grid cell having the centroid (x, y) ; $d_{x,y}^e$, $d_{x,y}^s$, and $d_{x,y}^r$ = the distance from the centroid of a grid cell to the nearest transmission line, picnic/camp site,

and major road, respectively; and $cc_{x,y}$ = the canopy cover ratio of a grid cell. It should be noted that the non-parametric smooth functions are deterministic across the study area once fitted by the historical wildfire events.

To estimate the models shown in Equations 1–3, historical wildfire data should be utilized. However, most of the voxels (i.e., each grid on a given day) do not have any historical fires due to fine spatiotemporal scale. For example, it is highly probable that there has been no fire in a specific square-kilometer grid on a given day (say September 1st). A subsampling method can be used to solve the issue and create a balanced subset by randomly sampling a ratio r of no-fire voxels to estimate the ignition probability P_{ign} (Brillinger et al., 2003; Preisler et al., 2004). The probability of an event that a fire occurred at the voxel k (E_A) conditioned on the event that voxel k was selected in the subset (E_B) can be expressed as:

$$P(E_A|E_B) = \frac{P(E_B|E_A)P(E_A)}{P(E_B)} \quad (4)$$

where $P(E_A|E_B)$ = the wildfire occurrence probability predicted by the balanced subset, which can also be expressed by p_r (Preisler et al., 2004):

$$P(E_A|E_B) = p_r = \frac{p_a}{p_a + (1 - p_a)r} \quad (5)$$

where p_a = the probability that a fire occurred at the voxel k . Equation (5) can be then written as Equation (6) through elementary algebra transformation (Preisler et al., 2004):

$$\logit(p_r) = \logit(p_a) - \log(r) \quad (6)$$

The estimation of the conditional probability that an ignition turns into a large fire $P_{\text{large|ign}}$ is straightforward and is assessed solely from the voxels associated with wildfire events. Therefore, applying the sub-sampling strategy to this conditional probability estimation, as used for ignition probabilities, is unnecessary in this case. It should be noted that the estimated wildfire occurrence probabilities are approximate, with an offset influenced by the subsampling ratio $\log(r)$, which accounts for the uncertainty arising from the sampling of data points. This uncertainty can be reduced by increasing the sub-sampling ratio r .

Utilizing the spatiotemporal occurrence probability of large wildfires, calculated through Equations 1–6, a set of representative wildfire scenarios can be generated. These scenarios are centered at the geographic grid clusters and the ERC percentile groups, thus encompassing a broad spectrum of plausible events. Geographic grid clusters, composed of closely situated grids, represent the spatial dimension of our analysis, while ERC percentile groups serve as clusters on the temporal scale. By combining these spatial and temporal clusters, this approach can balance the inherent uncertainty associated with wildfires while keeping computational demands manageable. The probability of a representative wildfire scenario, which combines a geographic grid cluster and an ERC percentile group, can be calculated by:

$$P_{\text{large}}(E, C) = 1 - \prod_{ERC \in E} \prod_{x,y \in C} (1 - P_{\text{large}}(ERC, x, y)) \quad (7)$$

where E = the ERC percentile group; and C = the geographic grid cluster.

In the second stage, the ignition locations of the representative wildfire scenarios are assumed to be the centroids of the geographic grid clusters. Wildfire growth is then simulated to generate spatiotemporal firefront and time-dependent fire map. Wildfire growth is mostly conditioned on topography as well as environmental and weather conditions. In this study, the Fire Area Simulator (FARSITE), a widely used wildfire simulation program now included in FlamMap 6, is used. FARSITE utilizes the Huygens principle (Richards, 1990), where small elliptical wavelets are formed around the perimeter of the fire (i.e., firefront) at the current time step, and the outer edge of these wavelets

becomes a new firefront at the next time step. Because an iterative simulation structure allows time-dependent inputs, FARSITE has the ability to model wildfire growth under heterogeneous weather, fuel, and topography conditions, which produces more accurate simulation results compared to the other software programs. However, FARSITE is computationally inefficient: FARSITE simulation time could be over 10 times longer than the other programs, which limits its usage when simulating a large and complex wildfire or generating a large suite of fires. On the other hand, other widely used wildfire simulation software programs, such as FlamMap, FSPro, and FSim, model wildfire growth based on the Minimum Travel Time (MTT) method developed by Finney (2002). MTT is a simplified raster-based fire growth model searching for the minimum travel time among the nodes of a grid. Fire shape is created by contouring the nodes with the same minimum travel time. Since this method holds weather and fuel conditions constant in time, its simulation results are not as precise as FARSITE. However, MTT greatly reduces simulation time. Since the prediction accuracy of fire growth at every time step is key to effective wildfire evacuation modeling, FARSITE is used in the fire growth modeling of this study despite its relatively high computational costs.

A set of grid-specific raster-based topographic inputs (e.g., elevation, slope, and aspect) and fuel property inputs (e.g., canopy cover, fuel mode, canopy stand height, canopy base height, and canopy bulk density) are collected and used as inputs for wildfire growth modeling. Additionally, hourly weather data including temperature, relative humidity, precipitation, wind speed, and wind direction are collected from the representative Remote Automated Weather Stations (RAWSS) in the study region to capture temporal characteristics. For each representative wildfire scenario, the spatiotemporal firefronts are generated and fed into the vulnerability assessment (Module 2) and evacuee response model (Module 3) to identify the locations of closed road segments and estimate the timing of evacuation trigger and departure time.

2.2. Module 2: Road network vulnerability assessment

Road networks play a pivotal role in facilitating evacuations, forming the foundation for transporting individuals to safe locations. Estimating the functionality states of road network components (e.g., highway bridges, road segments) during fires holds significant importance, as it directly influences evacuation efficiency. However, the vulnerability and functionality assessment of road network during wildfires has received minimal attention in existing research. While a number of researchers have constructed data-driven models for assessing the vulnerability of bridges to fire (Naser, 2021; Peris-Sayol et al., 2017), the scarcity of available data was a major issue: the data used in these studies have predominantly been sourced from news reports, existing literature, and bridge management authorities, which has caused the issue of limited data availability and restricted the broader applicability and robustness of these models. Similarly, experimental works in this area are still not sufficient to construct a fire fragility curve due to different laboratory settings and limited results. Moreover, these prior research initiatives have been limited to the vulnerability assessment of individual bridges and have not yet been extended to other types of components (e.g., road segments). These limitations have hindered a comprehensive understanding of how wildfires can impact transportation networks, necessitating further exploration and analysis.

Traditionally, roads have been perceived as constructed from non-combustible materials, largely resilient to physical damage in wildfire situations. However, the realities of wildfires introduce a series of practical challenges that extend beyond structural damage and have the potential to disrupt road networks significantly. These challenges include low visibility due to smoke, ashfall, flying embers, burning debris, falling trees, and landslides, which often lead to road capacity reduction or even road closures during wildfire events. In this module, a data-driven machine learning (ML) model for fire vulnerability assessment was created to estimate the probabilities of road closures resulting

Table 1

Selected major wildfire events, burned acres, and years in the western United States.

Fire	Acres	Year	Fire	Acres	Year
Dixie	963,405	2021	Camp	153,336	2018
Bootleg	413,717	2021	Rough	151,547	2015
Creek	379,883	2020	Archie Creek	131,596	2020
LNU Lightning Complex	363,220	2020	Dolan	124,310	2020
Claremont-Bear	318,777	2020	Woolsey	96,949	2018
Thomas	281,894	2017	Evans Canyon	75,693	2020
Carr	229,651	2018	Rocky	69,438	2015
Pearl hill	223,731	2020	Tamarack	67,054	2021
Monument	223,127	2021	Loyalton	56,477	2020
Caldor	221,786	2021	Archer Mountain	48,879	2017
Beachie Creek	193,566	2020	Earthstone	42,253	2017
Chetco Bar	191,242	2017	Slinkard	40,142	2017
Cold Spring	189,912	2020	S. Obenchain	32,671	2020
Range 12	176,610	2016	Bobcat	30,024	2020
Holiday Farm	173,393	2020	Mill	3,935	2022

from both direct and potential indirect impacts of wildfires (e.g., structural damage, visibility issues, safety concerns, obstructions, and roads reserved for emergency responders). The dataset was compiled from 30 major wildfire events that occurred between 2015 and 2022 in the western United States, as detailed in Table 1. The road closures during wildfires were identified using a combination of various public sources, including news reports and official announcements from agencies and local fire departments. Road segments reported as closed by these sources were labeled as closed roads in the dataset. All road locations and attributes were sourced from the OpenStreetMap (OSM) road layers. The selected wildfire events affected 18,581 road segments, with 1,427 segments experiencing closures during the events. To address the issue of the imbalanced dataset, the closed road segments were up-sampled to match the number of non-closed road segments, resulting in a balanced dataset containing 37,162 road segments. The up-sampling approach was employed to increase the representation of the minority class, a crucial step in training robust models on imbalanced datasets, which helps prevent the predictive model from overwhelmingly classifying instances into the majority group. It is important to emphasize that this up-sampling technique is a common practice and preserves the original data structure, focusing on giving equal weight to both the majority and minority groups.

The road properties considered in constructing the road closure predictive model included road type, whether it is a one-way road, its maximum speed limit, the number of intersecting roads, and the presence of bridges or tunnels on the segment. These road properties can be expanded to include additional detailed characteristics collected by federal or local agencies, allowing for customization to meet specific needs. Various ML models were adopted and tested to construct the road closure prediction models. These included naïve Bayes classifier (NB), k-nearest neighbors (KNN), support vector machine (SVM), random forest (RF), decision tree (DT), and neural network (NN). These ML models have been employed to perform various types of classifications and demonstrated high accuracy compared to traditional statistical models (Katzilieris et al., 2022; Xu et al., 2023). For illustrative purposes, this study used the default hyperparameter settings for the aforementioned ML algorithms, evaluating and comparing them based on prediction accuracy in the test dataset. Among various ML models, the Random Forest (RF) model was selected as the road closure prediction model due to its superior accuracy (97 %), significantly higher than the 70.5 % accuracy of traditional logistic regression, for example. The pre-trained RF model based on the up-sampled dataset was then applied to the road segments exposed to the representative wildfire scenarios to identify potential road closures.

2.3. Module 3: Evacuee response model

In the fourth module, traffic demand will be estimated at the individual evacuee, household, or vehicle level. Thus, in this module, individual responses and behaviors during evacuation are modeled and simulated. Similar to the classic four-step transportation planning model consisting of demand estimation, destination choice, mode choice, and trip assignment, evacuation modeling generally includes evacuation decision, traffic mode split, destination choice, departure time estimation, and information update procedure. Individual evacuees start to evacuate at different points in time even when they are exposed to the same level of wildfire threat (Katzilieris et al., 2022). Evacuation behavior is a key factor in realistically simulating evacuation at both the individual and community level and can be defined by a set of actions of individuals and external stimuli during an evacuation (D'Orazio et al., 2014).

To mathematically model an individual's evacuation behavior, a series of three events is considered in this study: evacuation trigger, decision time delay, and preparation time (Mas et al., 2012). First, four types of external stimuli serve as evacuation triggers for evacuees. An evacuation alert (Level I) lets people know that a wildfire threat is in their areas and suggests they consider evacuation in the event that it becomes necessary. An evacuation warning (Level II) indicates the high probability of a need to evacuate. An evacuation order (Level III) asks people to evacuate within a specified time period. A large portion of people may respond to these three levels of evacuation, but some people may wait until a firefront is within visible range of their initial locations (Mas et al., 2012). As such, evacuation triggers, which can actually induce an individual evacuee's response, may depend on his or her risk-averse attitude and other characteristics and should be identified and explicitly incorporated into evacuation modeling. The timing of each evacuation trigger can be estimated using the spatiotemporal firefront obtained from Module 1. For example, if the distance between the firefront and the initial location of an evacuee is within a specific threshold, one of the stimuli is activated, and the evacuee's response is simulated based on his/her risk attitudes and characteristics through Monte Carlo simulation (MCS). The departure time is not immediately after the timing of the evacuation trigger but may be delayed due to decision delay and preparation. Decision delay may occur if household members (or a group of people who plan to evacuate together) need to make a collaborative decision to evacuate. Decision delay may also depend on an evacuee's awareness, beliefs, and priorities (Beloglazov et al., 2016). Preparation time is also an evacuee-specific parameter.

To collect empirical data and understand highly unpredictable human behaviors during a wildfire evacuation, we conducted a web-based stated preference survey of residents in California, Oregon, and Colorado (Ma & Lee, 2024). These states were chosen because they experienced major wildfire events and evacuations in the past several years. Participants were recruited through Amazon Mechanical Turk (MTurk) between April and May 2022. The inclusion criteria for participating in the survey were residents living at the study site who were at least 18 years of age. A total of 1,312 participants completed the survey, but 554 responses were considered invalid because they failed to meet at least one of the validation checks: (a) attention check; (b) consistency check; (c) logical check; and (d) IP address check (Kennedy et al., 2020; Ma & Lee, 2024; Owens & Hawkins, 2019). Consequently, the final dataset had 758 valid responses. The online survey comprised a series of closed-ended quantitative and qualitative questions aimed at gathering detailed information about the respondents. These questions covered various aspects, including demographics, property and car ownership, mobility issues, risk perception, prior experience with natural hazards, and potential evacuation-related actions. It should be noted that the stated preference survey conducted in this study did not include any specific control strategies for respondents with prior evacuation experience. Further research should consider incorporating more targeted control strategies or conducting a revealed preference survey to

reduce potential biases and improve the specificity of the results.

Based on the survey responses, we constructed predictive models for evacuation triggers, evacuation destinations, and navigation uses during an evacuation using both traditional logistic regression models and ML models. While ML techniques have been widely used in evacuation modeling, their use in the context of wildfire evacuation has been limited, with only a few notable exceptions (Katzilieris et al., 2022; Xu et al., 2023). ML approaches offer distinct advantages over traditional statistical methods, particularly in handling complex relationships and nonlinear interactions between variables. Thus, in this study, we compared the prediction accuracies of traditional logistic regression models with various ML models (e.g., DT, RF, SVM, NB, KNN, NN) and chose the models exhibiting the highest prediction accuracies. A detailed comparison of the prediction accuracies of these models can be found in Appendix C. Specifically, RF classifier with small trees (max depth = 5), the RF classifier with large trees (max depth = 15), and the SVM with Gaussian kernel were chosen for the predictive models for evacuation triggers, evacuation destinations, and navigation uses, respectively. The superior performance of ML models compared to traditional logistic models observed in this study aligns with their demonstrated effectiveness in predicting evacuation preferences in Katzilieris et al. (2022) and Xu et al. (2023). On the other hand, for the evacuation preparation time, both statistical analyses and ML models showed low performance, with accuracies hovering around 28 %, only marginally better than pure random guessing. Consequently, we opted to sample the evacuation preparation times from the frequency distribution corresponding to each evacuation timing group (i.e., immediate, voluntary, and mandatory groups). A comprehensive description of the survey data collection process, analytical methods, discussions, and detailed descriptions of the predictive models can be found in Ma & Lee (2024).

2.4. Module 4: Agent-based traffic simulation

The reduced functionality of a transportation network (Module 2) can endanger human lives when combined with the elevated travel demand patterns (Module 3) during an evacuation. Module 4 will perform traffic simulation by combining Module 2 results with Module 3 results to predict traffic conditions during an evacuation and identify the critical parts of the transportation network for pre-fire risk mitigation actions. Travel demand modeling for wildfire evacuation can be generally classified into trip-based and activity-based approaches (Intini et al., 2019). The reference unit for the trip-based approach is origin–destination pairs, and the total demand is estimated at the aggregated level. This approach is usually adopted in macroscopic traffic simulation and is more suitable for evacuations involving large geographic areas. On the other hand, the activity-based approach modeling each evacuee as an agent is often adopted in microscopic traffic simulation and is more appropriate for regional evacuation modeling due to its high computational burden. Considering that wildfire evacuation is often limited to the community or county level, activity-based microscopic traffic simulation has been widely used in existing wildfire evacuation studies (Cova & Johnson, 2002; Grajdura et al., 2020; Mancheva et al., 2019). In this study, an activity-based microscopic traffic simulation is integrated within ABM. Two simple activity chains were assumed for households subject to evacuation orders. In cases where some household members are at the exposed property with vehicles available on-site, they enter the network directly and travel to evacuation exits or public shelters. Conversely, if no vehicles are available at the exposed property but another household member is at a work site with a vehicle, the working member first returns to the household to gather the remaining members before proceeding to the evacuation exits or public shelters.

There are various ABM software programs that have been used in evacuation studies. NetLogo, a java-based program, was used by Grajdura et al. (2020) and Siam et al. (2022) to model wildfire evacuations. Cova and Johnson (2002) used Paramics, which is a commercial traffic

simulation software, in simulating wildfire evacuation. Transportation Analysis and Simulation System (TRANSIMS) and Simulation of Urban Mobility (SUMO) are also widely used software programs in evacuation planning and simulation (Beloglazov et al., 2016; Lopez et al., 2018; Yin et al., 2014). SUMO is chosen to perform traffic simulation in the fourth module of this framework due to its accessibility and flexibility.

Each evacuee in the community is modeled as an agent, and based on the survey results obtained from Module 3, individual characteristics and decision rules are assigned to each agent. In the ABM, individual agents determine their actual departure time, evacuation route, and destination prior to departure. After the agents depart from their point of origin (i.e., the results from Module 3), they will enter the transportation network and choose their evacuation routes using their routing selection method (i.e., the results from Module 3). Some agents who will not use real-time maps may not be able to select the optimal evacuation routes because they will not have any real-time information, such as the locations of closed roads due to fire, traffic congestion, etc. As such, the spatiotemporal movements of individual agents are recorded at every time step.

The traffic carrying capacities of the links (i.e., the results from Module 2) are also dynamically updated as fire propagates over time. Dynamic updates on link capacity and individual agents' locations are incorporated into microscopic traffic simulation to predict traffic flows at the level of individual agents and capture traffic dynamics considering collective actions among the agents. The ABM framework for wildfire evacuation in damaged transportation settings provides time-dependent traffic maps to identify the critical parts of the transportation network that are the most vulnerable to wildfires and have great potential for causing traffic congestion during an evacuation. Moreover, the total number of evacuating agents during a given time is also obtained as a final result.

3. Case study: The City of Santa Clarita, California

3.1. Study site

To illustrate the application of the proposed ABM framework to a real-world community, the City of Santa Clarita, California was selected as a case study site. The City of Santa Clarita is located in northwestern Los Angeles County in California as shown in Fig. 2, and its 2020 population was 228,673. This city was chosen because many homes in Santa Clarita are located within WUI and at high wildfire risk. This positioning makes the city particularly susceptible to mass wildfire evacuations. For example, the Rye Fire, which ignited on December 5th, 2017, threatened over 1,000 homes in the city and prompted evacuation orders for approximately 1,300 homes. The fire also necessitated the closure of both directions of Highway 5, one of the major thoroughfares in the area. Another significant incident is the Tick Fire, which broke out on October 24th, 2019. This fire posed a risk to over 10,000 structures in the city and led to the evacuation of about 40,000 residents. The scale and frequency of such events highlight the urgent need for effective evacuation planning.

3.2. Simulation procedure

3.2.1. Module 1 Procedure: Wildfire simulation

The first stage of the wildfire simulation module is to generate spatiotemporally varying fire occurrence probabilities in the study region as shown in Equations 1–6. The required inputs for the wildfire occurrence probability estimation models (see Equations 1–2) include gridded daily climate data (i.e., the ERC), gridded human factor data (i.e., population density, distances from grid centroid to the closest major roads, camp/picnic sites, and transmission lines), and gridded topographic and fuel layers. The University of Idaho Gridded Surface Meteorological Dataset (gridMET) (Abatzoglou, 2012), daily high-spatial resolution (~4 km) climate data collected since 1979, was used

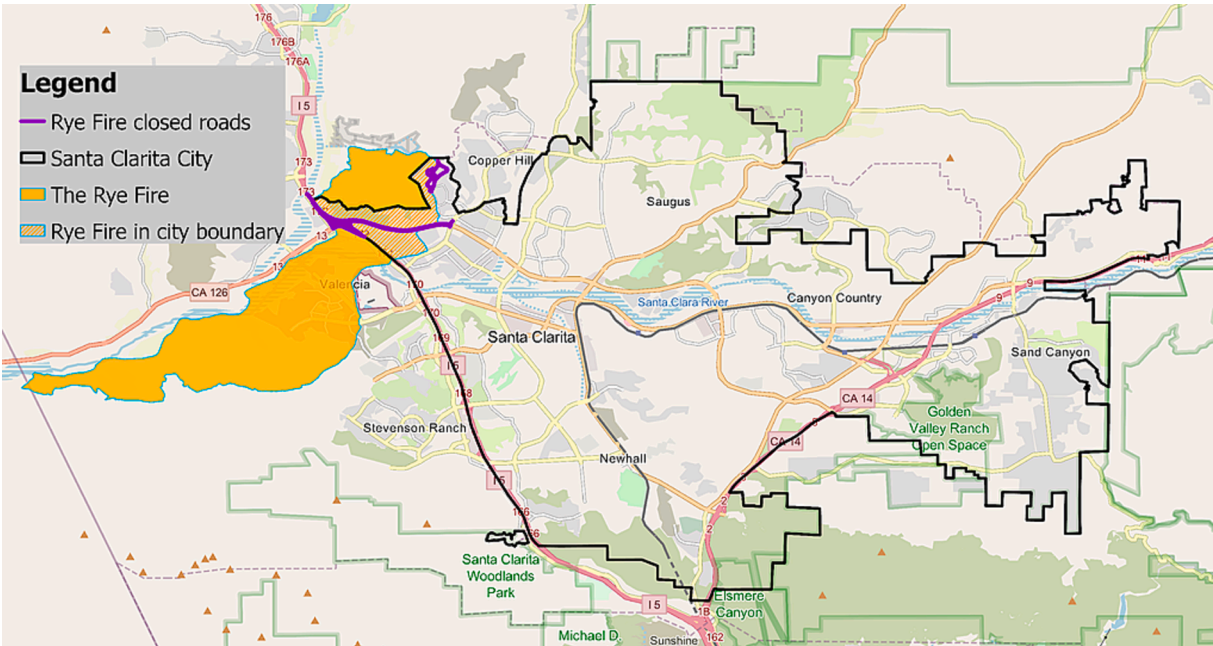


Fig. 2. The City of Santa Clarita affected by the Rye Fire.

for the climate data. The gridded human factor data were generated by utilizing OSM data. The gridded topographic and fuel data available from the LANDFIRE program had a spatial resolution of 30 m and were used as the input. Given that the largest wildfire in California history scorched more than 1 million acres, historical wildfire events within 31 zip codes surrounding the city, encompassing approximately 2,739,905 acres, were collected for estimating wildfire ignition probabilities. During the period between 2002 and 2021, there were 5,288,820 voxels. Among them, only 435 voxels had historical fire events that burned areas greater than 100 acres, while the remaining 5,288,385 voxels did not have any large fires. As described in Section 2.1, to address the issues with imbalanced data, 435 voxels were first randomly sampled from the no-fire set, which is a subsampling method. Then, the subset consisting of 435 fire voxels and 435 no-fire voxels was divided into training and testing sets to construct the wildfire occurrence probability estimation model. The accuracy of the ignition probability model yielded 80 % prediction accuracy, while the conditional probability of an ignition

Table 2
The probabilities of the representative wildfire scenarios (i.e., $P_{large}(E, C)$).

	0th – 40th	40th – 60th	60th – 80th	80th – 90th	90th – 97th	97th – 100th
1	0.61 %	0.34 %	0.52 %	0.23 %	0.26 %	0.65 %
2	0.15 %	0.09 %	0.14 %	0.06 %	0.07 %	0.18 %
3	0.24 %	0.14 %	0.22 %	0.10 %	0.11 %	0.28 %
4	0.34 %	0.18 %	0.28 %	0.12 %	0.14 %	0.34 %
5	0.43 %	0.23 %	0.35 %	0.16 %	0.18 %	0.44 %
6	0.32 %	0.18 %	0.28 %	0.12 %	0.14 %	0.35 %

turning into a large fire model yielded 75.2 % accuracy. The number of fire voxels was relatively small because this study focused only on the areas surrounding Santa Clarita City and utilized only a 20-year historical record. This may have contributed to the relatively low prediction accuracy. There are multiple ways to improve its accuracy, such as using longer climate data (e.g., collected during the period between 1979 and 2021).

To generate a set of representative wildfire scenarios, the region encompassing and surrounding the City of Santa Clarita was divided into 24 grid cells, each measuring 4 km by 4 km, covering a total of 94,888 acres. More specifically, the region was divided into six clusters (see Fig. 3), with each cluster comprising four grid cells. The red dots denote the centroid of each cluster, and the numbers indicate the labels of the clusters used in Table 2. To account for temporal variability, six ERC percentile groups were considered: 0th–40th percentile, 40th–60th percentile, 60th–80th percentile, 80th–90th percentile, 90th–97th percentile, and 97th–100th percentile. For the spatial centroid of each cluster and each ERC percentile group, a representative wildfire scenario was generated. Table 2 presents the total 36 representative scenarios and their probabilities calculated based on the spatiotemporal occurrence probability map.

The ignition location of each representative wildfire scenario was designated as the spatial centroid of the respective cluster. The required inputs to FARSITE include hourly weather data and a set of raster files (i.e., elevation, slope, aspect, fuel model, canopy cover, stand height, canopy base height, and canopy bulk density). The hourly weather data were available from the RAWSSs located throughout the United States, while all raster files were downloaded from the LANDFIRE program. It

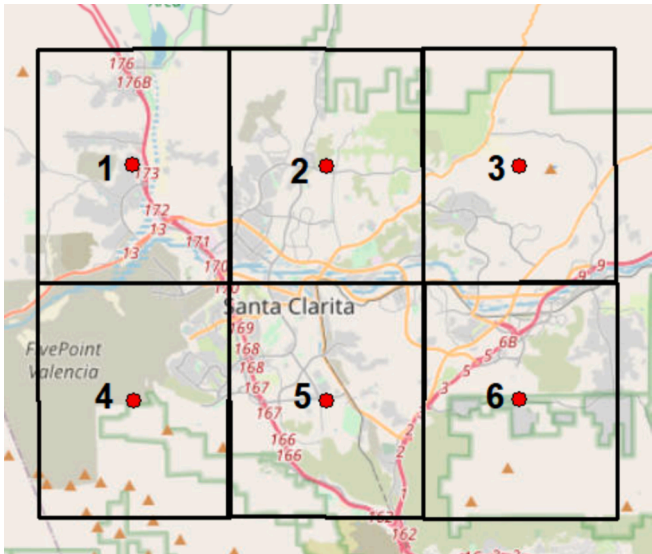


Fig. 3. Clusters and their centroids.

should be noted that the hourly data used to simulate each representative wildfire scenario was derived from historical weather records. The specific date for the hourly weather data was selected by identifying a date when the ERC matched the representative ERC within its percentile group. This method ensures that the weather conditions used in the simulation accurately reflect the conditions corresponding to the representative ERC, thereby enhancing the reliability of the realistic wildfire scenario simulations. Moreover, the 40 Scott and Burgan Fire Behavior Fuel Models (FBFM40) do not support fire spread into urban and suburban areas, which is a limitation that contradicts historical wildfire events in the Santa Clarita area (e.g., Tick Fire, Rye Fire). To address this discrepancy in the case study, we replaced the urban and suburban development type (NB1) with the short, sparse dry climate grass type (GR1) to allow fires to propagate into urban and suburban areas.

For each of the representative scenarios, we assumed that fire initiation occurred at 9 am and stopped propagating at 8 pm. In other words, the conclusion of the simulation does not necessarily indicate that the fire has been completely contained. This assumption is reasonable because, in many real-world cases, evacuation orders are lifted within a day, reflecting the immediate need for evacuation during the most critical period of wildfire spread. Moreover, the accuracy of the FARSITE simulation is projected to diminish after the initial period due to substantial uncertainties introduced by human intervention. Firefighting strategies, weather changes, and other dynamic factors can significantly alter the fire's behavior, making long-term simulations less reliable. By constraining the simulation to this timeframe, we can focus on the critical early stages of wildfire spread and evacuation, which are crucial for understanding and improving network performance and evacuation dynamics under urgent conditions.

To test the FARSITE wildfire simulation performance, the simulated fire perimeter during the first burning day of the Rye Fire (December 5, 2017) was compared with the actual perimeter, as shown in Fig. 4. Only the first-day simulated fire perimeter was considered herein because the evacuation order was lifted the evening of the first burning day, and further wildfire growth simulation was unnecessary for estimating

evacuation. Moreover, a lack of a suppression model in FARSITE required additional assumptions about human suppression efforts in order to stop the fire. While FARSITE has limitations in simulating the Rye Fire from its ignition to full containment, the similar propagation direction of both simulated and actual fires (see Fig. 4) indicates that the model has the capability to simulate fire growth successfully until human intervention gets involved.

3.2.2. Module 2 Procedure: Road network vulnerability assessment

The locations of the roads exposed to the representative wildfire events are illustrated in Fig. 5 as black links. The attributes of the roads were extracted from the OSM road layer, including road type, road access restrictions (one-way or both ways), maximum speed limit, and vertical relationships between crossing or overlapping roads, bridges, and tunnels. These attributes were processed and utilized in the road network vulnerability assessment model, which was estimated based on historical closures described in Section 2.2. Out of the 3,623 road segments affected by the representative events, 54 road segments were predicted to be closed, shown as the red segments in Fig. 5. The reduced capacities of the road segments for each representative wildfire event were then integrated into the traffic simulation in Module 4.

3.2.3. Module 3 Procedure: Evacuee response model

In the third module of the proposed framework, the evacuee response model was simulated to reflect diverse individual behaviors and responses during an evacuation. To apply the evacuee response model developed based on the survey data, detailed information about the properties and characteristics of individual residents and households in the community was necessary. Whereas demographic information was available from the Census dataset (U.S. Census Bureau, 2021), some of the key explanatory variables needed to estimate their evacuation timing or the use of real-time navigation (e.g., risk attitude, wildfire evacuation experience) were often not available. Thus, in this study, we generated a synthetic population for the City of Santa Clarita.

Two standard approaches to generating a synthetic population are synthetic reconstruction (SR) and hill climbing (HC) (Namazi-Rad et al.,

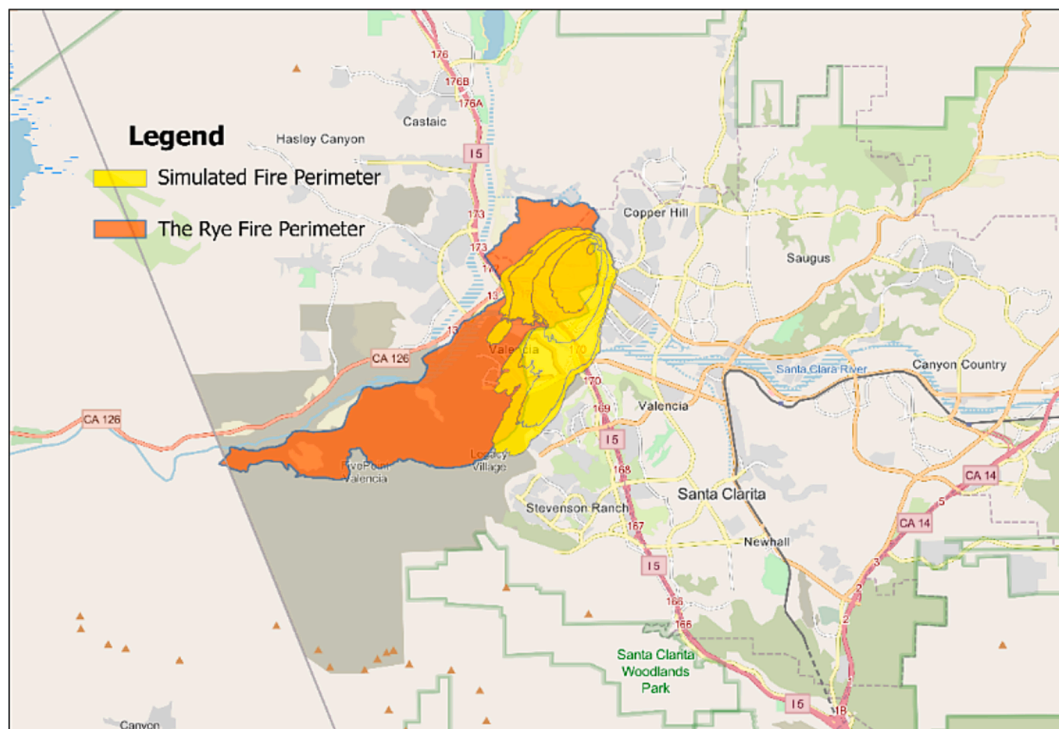


Fig. 4. Comparison between the simulated fire perimeter (only during the first burning day) and the Rye Fire perimeter.

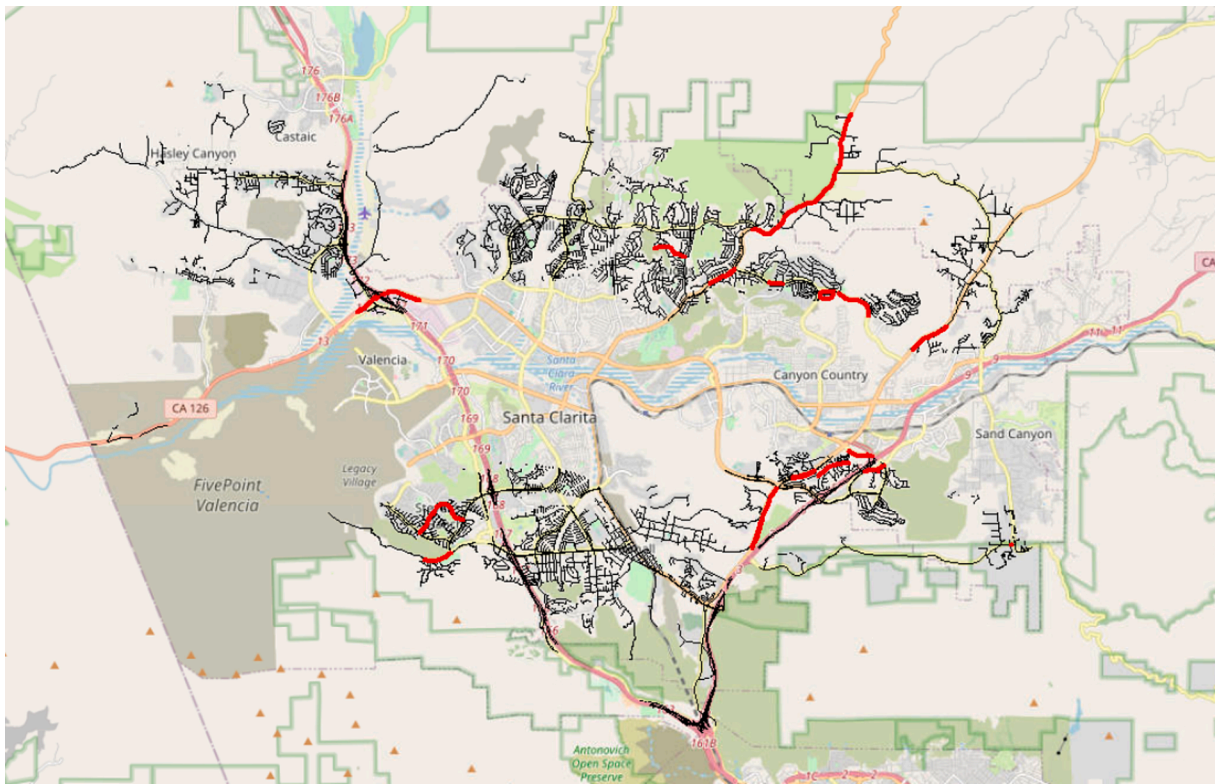


Fig. 5. Locations and predicted closures of road segments due to the representative wildfire scenarios (black: roads exposed to the representative wildfire scenarios; red: closed roads). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2014). The SR approach is a traditional method for generating a synthetic population based on both disaggregated- and aggregated-level data. By assuming that disaggregated-level data (i.e., seed data) can fully represent the target population, this approach samples individuals with the required sociodemographic characteristics from the true population in the study area, which is typically extracted from Census data, using a weighting technique. The HC approach is an optimization algorithm that maximizes an objective function. It starts with a random selection of units from the seed population and iteratively replaces them until improvement becomes negligible (Namazi-Rad et al., 2014). While both approaches are capable of generating reliable synthetic populations relative to the true population's marginal distributions (Jain et al., 2015), the SR approach was adopted in this study because user-friendly and well-developed software programs were available.

First, we used PopGen to generate a synthetic population in the City of Santa Clarita at the census tract level. PopGen is a software program that produces synthetic populations while controlling and matching both individual- and household-level attribute distributions (Bar-Gera et al., 2009; Konduri et al., 2016; Ye et al. 2009). The seed population was adopted from the Public Use Microdata Sample (PUMS) produced by the U.S. Census Bureau, which was a small portion of the true population (including both households and individuals) without identification. The marginal distributions of the control population characteristics in the study area were available from the Census Bureau's American Community Survey (ACS). The generated synthetic population consisted of 83,558 households and 246,830 residents with no significant difference from the marginal distributions of the true population's key characteristics (e.g., household size, household income, the presence of children, age, gender, race, education, and employment status). All individuals in the synthetic population were given a unique ID number so that the individuals having the same ID number were assumed to be in the same household.

To relate each household to its home, the locations, shapes, and dimensions of all buildings in the City of Santa Clarita were obtained from

the OSM database. Buildings were classified into residential buildings and commercial buildings. The residential buildings having the top 30 % shape length were labeled as multifamily residential buildings. The ratio of 30 % was determined based on the 2019 U.S. Census Bureau American Housing Survey that 31.4 % of housing in the U.S. were multifamily housing. For each census tract, all synthetic households in the census tract were randomly assigned to residential buildings in the same census tract. Similarly, all individuals who worked full-time or part-time were randomly assigned to workplaces (i.e., commercial buildings). It should be noted that the assumed land use types (e.g., single-family or multifamily residential properties) derived from the OSM database can be replaced with more detailed and accurate parcel-level data. Parcel data, often maintained and provided by local planning or zoning departments, includes comprehensive information on property boundaries, land use classifications, and ownership details.

After allocating individuals to their households, homes, and workplaces, it was assumed that each household formed a decision group and shared the same evacuation decision during the evacuation process. Then, the evacuee response modeling for each household was constructed based on the survey data and models described in Section 2.3. The evacuation timings of all synthetic households in the city were determined based on the RF model (see Appendix C). The explanatory variables for evacuation timing included education, insured status, routine use of navigation, past effective evacuation experience, disaster plan, and financial risk preference. Whereas the first two characteristics of synthetic households were already realized during the synthetic population generation, the last four properties of each household were not known because these properties were often not available from the U. S. Census data. Thus, using MCS, these properties of each household were randomly sampled from the histograms of the survey results (see Appendix A). Additionally, some of these explanatory variables are individual-level variables (e.g., education and routine use of real-time navigation). This study assumed that the survey respondent was representative of his or her household or primary decision-maker and used

the individual-level variables to predict household-level evacuation decisions. However, it should be noted that this assumption may introduce bias due to representative issues (Hung & Wang, 2022; Seebauer et al., 2017).

The results indicated the following evacuation intentions among synthetic households: 21.1 % intended to evacuate without an official order; 34.2 % planned to evacuate under a voluntary evacuation order; 39.7 % intended to evacuate under a mandatory evacuation order; and 5 % chose to stay or evacuate only after observing physical cues, such as embers, smokes, or flames. In combination with the wildfire growth simulation results from Module 1, the evacuation times of all households were determined. At every time step, once a new firefront was updated, three buffer zones were established to generate three types of evacuation triggers: a 2-mile buffer around the updated firefront was used to trigger the evacuation of the immediate evacuation group (those who evacuate without an official order); a 1-mile buffer triggered the evacuation of the voluntary evacuation group; and a 0.5-mile buffer initiated the evacuation of the mandatory evacuation group. The actual departure time could be delayed due to decision delay and preparation. While the time associated with evacuation delay can be affected by various evacuee- and household-specific parameters (e.g., awareness, risk attitudes, priorities, and beliefs), it is difficult to identify key independent factors affecting the delay time and their statistical relationship because of the abstract nature of major parameters. Thus, in this study, the delay time of each household mainly due to evacuation preparation was randomly sampled from the frequency distribution of preparation time (Ma & Lee, 2024).

The evacuation destinations of the synthetic households are also uncertain. As presented in Appendix B, households have a number of destination choices, including someone else's house, hotel, public shelter, secondary residence, portable vehicle, and rental house. In this case study, it was assumed that public shelters were located in the City of Santa Clarita, while other destinations were located outside the city. Fig. 6 shows the three schools that are located within the city. These schools were considered public shelters in the case study. The intention

to evacuate to the public shelter was evaluated based on the regression model (Ma & Lee, 2024) in which the explanatory variables included age, gender, household size, home ownership, the presence of children, car ownership, routine use of navigation, insured status, past effective evacuation experience, evacuation plan, confidence in property, and risk preferences for personal safety and financial safety. The regression results indicated that 362 households (0.4 %) intended to evacuate to public shelters. Although this percentage seems to be small, this finding aligns with real-world observations of low public shelter usage: for example, during the Rye Fire event, an estimated 2 % of evacuees used public shelters. Despite their limited use, public shelters remain vital in emergency preparedness and response, offering a critical safety net for individuals and families without alternative evacuation options. Thus, the households predicted to evacuate to public shelters were randomly assigned to one of the three shelters. For households whose final destinations were not public shelters, three major community exits, as shown in Fig. 6, were assigned as their evacuation routes.

In addition to simulating evacuation timing and destination preferences, this module also examined individuals' inclinations toward utilizing GPS navigation during evacuations. The adoption of GPS navigation can facilitate access to real-time travel information, enabling individuals to dynamically adjust their routes to avoid traffic congestion while evacuating. The use of GPS navigation by individual households was predicted using the SVM model (Ma & Lee, 2024). This model incorporated a range of explanatory variables, including employment status, property ownership, the presence of household members with special needs, routine navigation usage, past evacuation experience, and safety risk attitudes. Among the 83,558 synthetic households, the SVM model projected that 10,971 households would opt for GPS navigation during the evacuation process, constituting approximately 13.1 % of the total population.

This module generated the evacuation timing, departure time, GPS navigation usage, and final destinations for all synthetic households based on the survey results. Detailed descriptions of the survey and predictive models can be found in Ma & Lee (2024). However, public

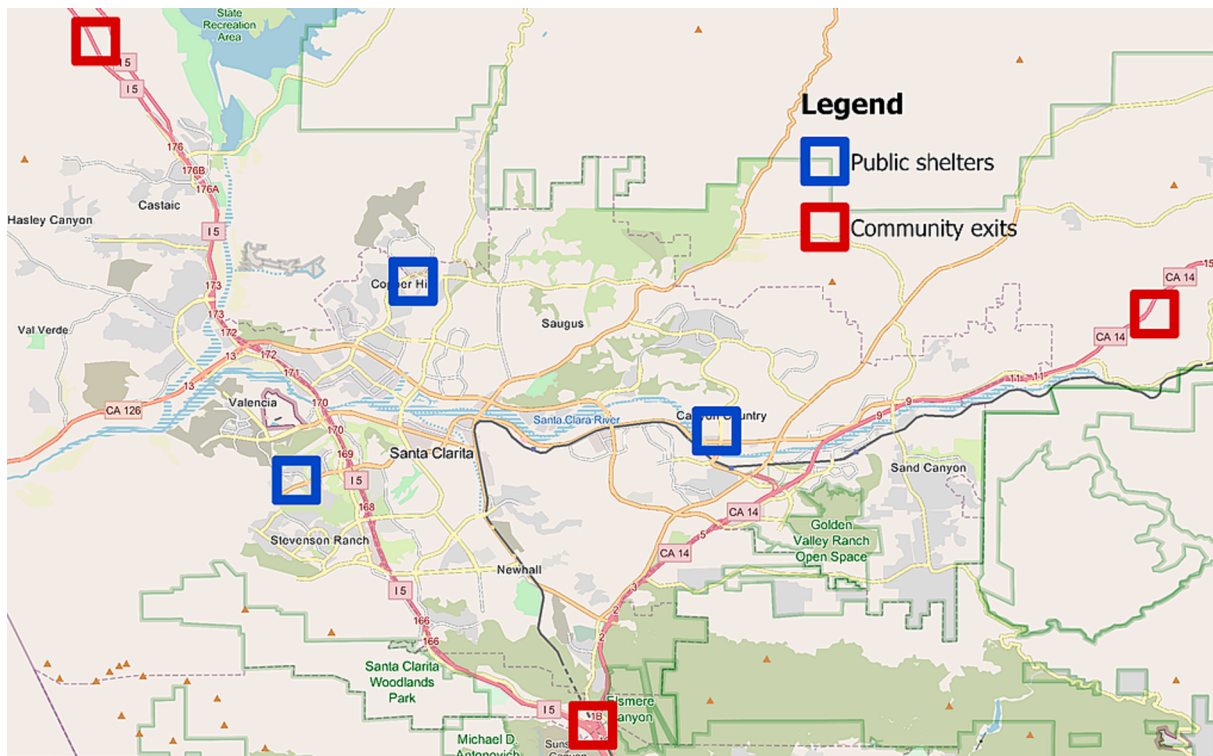


Fig. 6. The locations of public shelters and community exits.

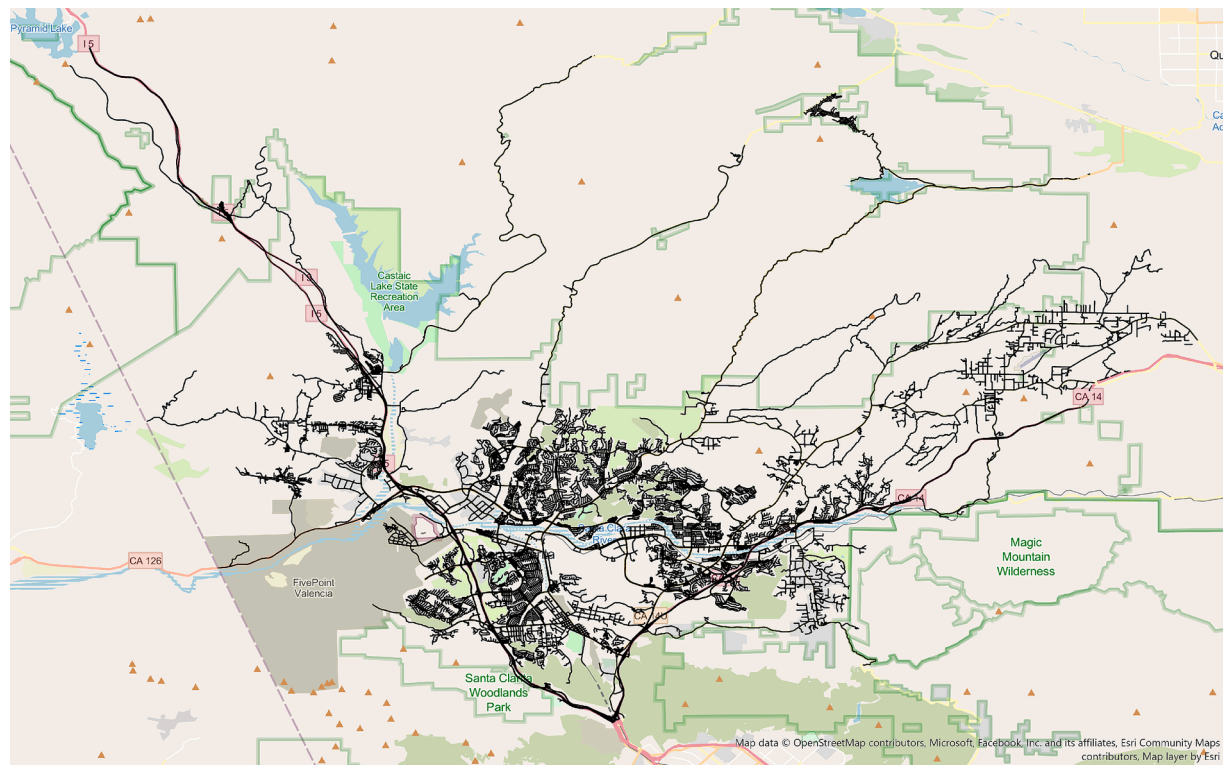


Fig. 7. Vehicle-only transportation network in the City of Santa Clarita.

transportation was not considered in this module. Thus, we assumed that households without vehicle ownership, yet desiring to evacuate, would utilize hypothetical vehicles for the evacuation process. As this assumption applied to only 4,142 (5 %) of the synthetic households, its

impact on the overall number of vehicles within the network and on simulation performance remains marginal. In future investigations, these hypothetical vehicles could potentially be replaced by public transportation or ride-sharing alternatives. Finally, a total of 166,069

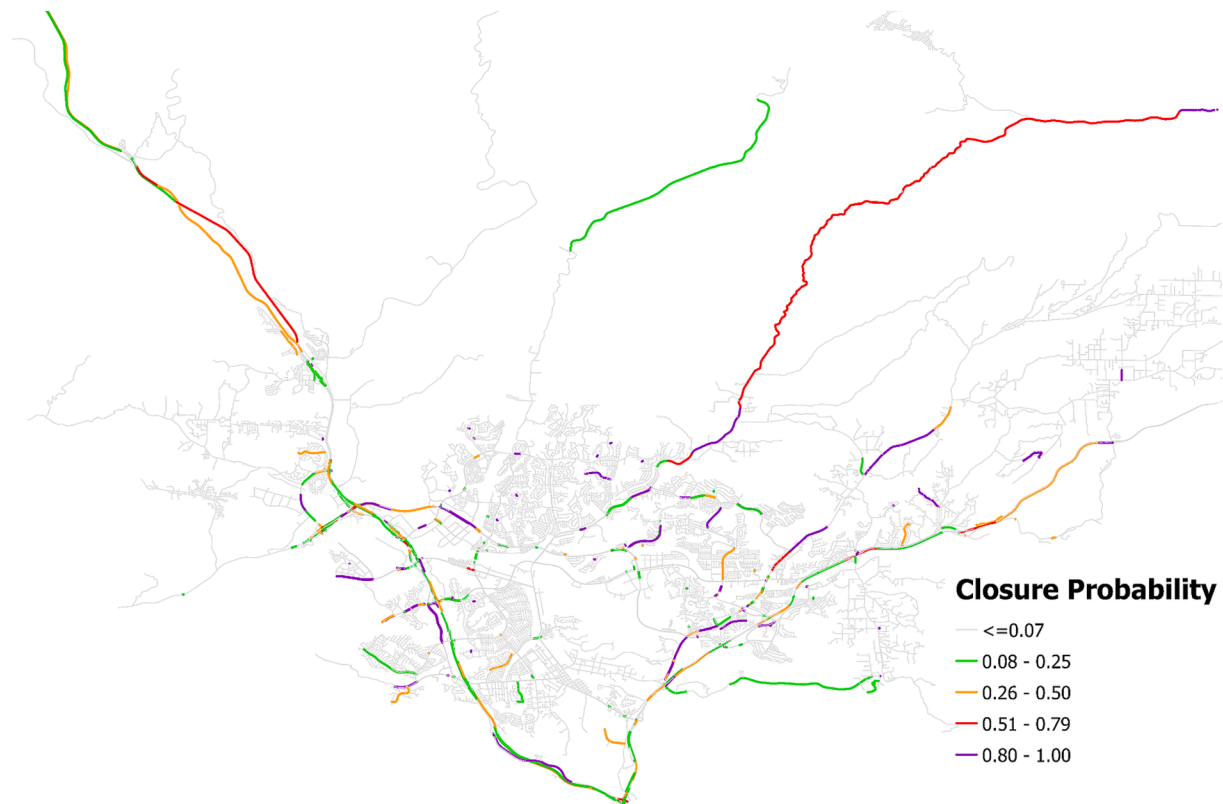


Fig. 8. Conditional road closure probability map in the City of Santa Clarita.

vehicles (both actual and hypothetical) were used during a wildfire evacuation. At each time increment, the module captured and documented the impending departures of vehicles, which was a crucial aspect factored into the estimation of traffic demand within Module 4.

3.2.4. Module 4 Procedure: Agent-based traffic simulation

The agent-based traffic simulation was performed using SUMO. The vehicle-only transportation network in the City of Santa Clarita is illustrated in Fig. 7. At every time step, the network capacity (i.e., the output from Module 2) and traffic demand (i.e., the output from Module 3) were updated and used as input data in SUMO. In SUMO, vehicles were considered autonomous agents, and their actions and interactions were simulated during a wildfire evacuation. Moreover, the evacuees were classified as navigation users and non-navigation users as described in Section 2.4. With the GPS navigation device, the navigation users could access the real-time map and reoptimize their route at every 180 simulation seconds and reroute if any faster routes were identified. The non-navigation users, on the other hand, only took the shortest (but not the fastest) routes because they chose familiar routes or relied on conventional maps during an evacuation without having real-time information. The results from this module were the locations of all vehicles at every second during the evacuation process, the time series of the number of vehicles in the transportation network, the number of vehicles that successfully evacuated, average speed, etc.

3.3. Results and discussions

After generating a set of representative wildfire scenarios for the City of Santa Clarita, we estimated the conditional closure probabilities of all road segments within the vehicle-only road network given that these segments are enclosed by wildfire perimeters (see Fig. 8). This estimation utilized the pre-trained road network vulnerability assessment model described in Section 2.2. The results reveal a pattern where residential and minor roads are less likely to be closed, whereas major roads and highways exhibit a higher likelihood of closures. It is not uncommon for wildfires to result in the closure of highways and major roads due to safety concerns. These closures may be partly caused by the direct and indirect impacts of wildfires, such as weakened pavement, landslides, debris, and dense smoke. Additionally, firefighting efforts can also contribute to the closure of major roads. For example, fire engines and other equipment may need to be stationed on or near these routes, and aerial firefighting operations, such as the dropping of water, require clearing the area to ensure safety. The conditional closure probability map, such as the one shown in Fig. 8, serves as a valuable resource for risk management officials, offering a holistic perspective on network functionality and accessibility under wildfire threats. By providing a pre-event outlook on potential road closures and highlighting the areas most vulnerable to road closures, this map empowers emergency managers to develop more effective evacuation routes that minimize reliance

on major roads and highways. This can help ensure smoother and safer evacuations during wildfire events. Moreover, it can allow for better allocations of resources, such as deploying firefighting units, traffic control personnel, and emergency responders to critical areas in advance, while informing decisions about infrastructure improvements (e.g., reinforcing major roads, creating alternative routes, and implementing firebreaks to reduce the risk of closures).

To illustrate both spatial and temporal variations in evacuation performance and better understand evacuation dynamics, Fig. 9 portrays the cumulative count of vehicles entering and evacuating the network over time. This cumulative number was averaged over the ERC percentiles for each grid cluster. Fig. 9(a) presents the varying average traffic inflows for each geographic grid cluster on a per-second basis. Since there are six representative wildfire scenarios for each cluster, the average traffic inflows represent the mean traffic inflows during wildfire scenarios that ignited within the respective cluster. As shown in Fig. 9(a), clusters C2 and C5 exhibit the highest volume of evacuations compared to the remaining clusters. This indicates that wildfires ignited in C2 or C5 have the potential to induce mass evacuations, which can lead to severe traffic congestion. Fig. 9(b) illustrates each cluster's mean cumulative evacuation rate. This rate is defined as the ratio between the cumulative number of vehicles that have successfully evacuated and exited the network and the total number of vehicles that departed. In clusters C1, C2, and C5, a large portion of vehicles successfully evacuated within the first 3 h after the fire ignited. This rapid increase in evacuation rates may indicate an effective initial response, suggesting that the evacuation routes in these clusters were capable of handling the initial surge of traffic to some extent. Additionally, residents in these clusters might be highly aware of the wildfire threat and promptly comply with evacuation orders. However, it should be noted that despite the effectiveness of the initial response, severe traffic congestion was observed in clusters C2 and C5 due to the high volume of evacuation traffic, which can be found in Fig. 9(a).

Fig. 10 illustrates the average cumulative time loss at every road segment during the first three hours following a fire outbreak. This three-hour time frame was selected because the most dramatic increase in traffic inflows was observed (e.g., Fig. 9(a)). The cumulative time loss is defined as the total number of seconds vehicles lost due to congestion, summed over all vehicles that traversed a particular road segment. Time loss is one of the built-in performance metrics in the SUMO simulation. It refers to the additional time a vehicle spends on its journey due to factors such as traffic congestion, traffic lights, and other delays. It represents the difference between the actual travel time and the ideal travel time if the vehicle were able to travel at its desired speed without any interruptions. A higher cumulative time loss value indicates that more vehicles experienced significant delays, suggesting that these segments are prone to severe congestion during wildfire evacuation. As shown in Fig. 10, the majority of road segments within the City of Santa Clarita are expected to encounter only minor slowdowns, with time delays

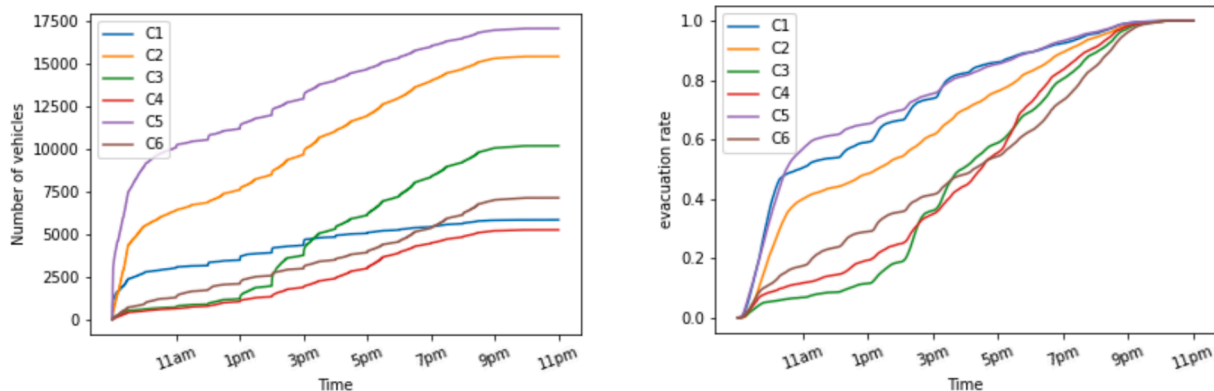


Fig. 9. (a) Mean cumulative traffic inflows for geographic grid clusters and (b) Mean cumulative evacuation rate for geographic grid clusters.



Fig. 10. Spatial distribution of average cumulative time loss on road segments (unit: seconds).

between 100 and 300 s. A subset of roads, particularly in the southern and northwestern sectors of the city (Clusters 1 and 5), may experience significant traffic congestion, resulting in time losses exceeding 5000 s. These areas of pronounced congestion are highlighted in blue and purple. According to the highlighted regions, severe traffic congestion is likely to occur at highway entry points partly because evacuees

generally prefer to use highways due to their higher capacity and speed. Such high traffic flows could impede traffic entering from the on-ramps, causing significant delays on these highway segments. Fig. 10 also identifies another vulnerable area in the residential zones located in the north of the city (Cluster 2). Residents in these areas may face heightened risk due to the limited evacuation routes available, increasing the



Fig. 11. Spatial distribution of annual expected cumulative time loss on road segments (unit: seconds).

likelihood of delayed evacuations during large-scale events. This spatial visualization of cumulative time loss offers emergency managers a valuable tool for pre-event assessment of network performance. By understanding which areas are most likely to experience severe congestion and delays, they can implement targeted strategies to improve evacuation efficiency and safety.

The spatial distribution of cumulative time loss can be further analyzed within a probabilistic framework and expressed as the annual expected time loss for each road segment. The annual expected time loss is a metric that quantifies the potential impact of wildfire evacuation scenarios on traffic performance by combining both the likelihood of various wildfire events and the severity of the resulting traffic delays. It is calculated as a weighted sum of the cumulative time loss for each representative wildfire scenario, with the weighting based on the probability of each scenario occurring within a year. This measure provides insight into how frequently road congestion might occur during evacuations and how severe the traffic delays could be. Fig. 11 illustrates this visual representation. Similar to Fig. 10, the analysis results in Fig. 11 indicate that the southern and northwestern regions of the City of Santa Clarita are likely to experience severe congestion during wildfire evacuations. These highly congested segments also align well with the road closure probability map shown in Fig. 8, indicating that road closures further exacerbate traffic congestion, especially when combined with high volumes of traffic inflow (Fig. 9(a)). Thus, these segments require targeted wildfire risk mitigation actions given their likelihood of closure and high traffic volume during evacuation events. However, it is important to note that most road segments experience negligible congestion in terms of annual expected time loss. This is because the calculation takes into account the annual probability of wildfires affecting the region. While a majority of these road segments may experience minor congestion when a wildfire does occur, the annual expected time loss provides a more comprehensive picture by reflecting how often such events are anticipated to impact the region. Rather than focusing on the absolute value of the annual expected time loss, it may be more beneficial to use these values in a relative manner to compare which road segments are more likely to experience congestion from a probabilistic standpoint. This approach can help planners and decision-makers prioritize mitigation efforts and resource allocation to those areas where the risk of significant congestion is not only high during wildfire events but also recurrently significant over time.

4. Concluding remarks

Effective community-based transportation evacuation planning is a critical concern for state and local policymakers. When wildfire risks to communities cannot be reduced by the first line of defense (e.g., wildfire countermeasures), evacuation becomes the essential second line of defense. Inadequate planning and underestimating the complexities of evacuation can lead to tragic human casualties during wildfire events. An evacuation simulation model may assist a well-developed evacuation plan and ultimately could save human lives.

This study proposed an ABM framework designed for wildfire evacuation simulation in damaged transportation settings. This novel approach integrates wildfire simulation, road network vulnerability assessment, evacuee response modeling, and traffic simulation, aimed at predicting traffic conditions during wildfire evacuation and identifying the critical segments of the road network that necessitate pre-fire risk mitigation measures. This ABM framework can generate outputs at both disaggregated and aggregated levels. At the disaggregated level, it can

provide detailed information such as individual vehicle locations and speeds at every second. At the aggregated level, it generates metrics including the cumulative count of departing or evacuating vehicles, the number of vehicles currently within the network or trapped in traffic, and time-series data depicting average travel speeds. This dual-level output capability allows for comprehensive analysis and insights into both fine-grained individual behaviors and broader traffic patterns during evacuation scenarios. Additionally, this proposed framework introduces the integration of damaged traffic settings into the evacuation process. This aspect underscores how the reduced network capacity due to wildfires affects evacuation efficiency, particularly when combined with heightened travel demand.

The main contributions of this study include (a) the incorporation of an advanced wildfire hazard modeling and vulnerability assessment to reflect realistic evacuation dynamics in damaged transportation settings, (b) the development of a more comprehensive evacuee response model based on a stated preference survey to predict individual evacuees' behaviors as a firefront approaches, and (c) the fully probabilistic assessment of evacuation performance. However, the study has a number of limitations that may hinder its accuracy. First, public transportation and ride-sharing behavior were not considered. Second, the evacuation destination was randomly assigned based on the survey results, whereas in the real world, evacuees may choose the closest shelter or exit. Third, the survey platform MTurk, utilized in this study, is less effective in accurately targeting and engaging individuals who have previously experienced wildfire evacuations, which could affect the reliability of insights. Fourth, while off-the-shelf simulation applications (e.g., FARSITE, SUMO) were used in this study for wildfire and traffic simulations, they could be replaced with other state-of-the-art models for improved performance. To further enhance the accuracy of the ABM framework and broaden its applicability, future works will be conducted, including (a) the introduction of public transportation, (b) the incorporation of carpool behavior to remove hypothetical vehicles, (c) the use of more focused surveys targeting individuals experiencing wildfire evacuations, and (d) the development of a more detailed simulation model for assigning evacuees to the closest shelter or exit.

CRedit authorship contribution statement

Fangjiao Ma: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Ji Yun Lee:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A: Survey data on individual and household characteristics

Characteristics	Classes	Frequency
Gender	Female	47.2 %
	Male	52.8 %
Age (years)	18 – 29	23.1 %
	30 – 39	35.5 %
	40 – 49	20.2 %
	50 – 59	13.3 %
	60 – 69	6.8 %
	>= 70	1.1 %
Education	Less than a high school diploma	0.6 %
	High school diploma or equivalent	15.4 %
	Associate degree	10.0 %
	Bachelor's degree	58.5 %
	Graduate degree	15.5 %
Ethnicity	Caucasian	68.4 %
	Asian	12.0 %
	African American	4.9 %
	American Indian, Alaska Native, or Native Hawaiian	1.4 %
	Two or more races	5.7 %
	Other races	7.6 %
Annual household income before taxes	Less than \$49,999	28.6 %
	\$50,000 to \$99,999	47.9 %
	\$100,000 to \$199,999	17.6 %
	Over \$200,000	5.9 %
Employment	Employed, working 40 or more hours per week	71.1 %
	Employed, working less than 40 h per week	16.3 %
	Not employed	5.5 %
	Retired	2.6 %
	Disabled, not able to work	1.0 %
	Others	3.4 %
Mode of commuting	Personally owned car, truck, or van	63.4 %
	Personally owned motorcycle	10.2 %
	Public transportation (e.g., bus, streetcar subway)	5.7 %
	Carpool	0.7 %
	Shared rides (e.g., Uber, Lyft)	1.5 %
	Taxicab	1.5 %
	Personally owned bicycle	1.5 %
	Walk	2.1 %
	Work from home	10.9 %
Decision-making role	Other	2.5 %
	Primary decision-maker	23.8 %
	Sole decision-maker	44.5 %
Family Size	Share decision-making roles equally with others	31.7 %
	1	16.3 %
	2	19.2 %
	3	21.4 %
	4	32.6 %
Property ownership	5 or more	10.5 %
	Own without mortgage	24.8 %
	Own with mortgage	37.8 %
	Rent	35.0 %
Presence of children	Other	2.4 %
	Yes	54.5 %
Presence of elderly	No	45.5 %
	Yes	26.1 %
Special need required	No	73.9 %
	Yes	25.4 %
Number of vehicles owned	No	9.5 %
	1	57.2 %
	2	23.9 %
	3 or more	9.4 %

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Characteristics	Classes	Frequency
Length of residency on the current property	Less than 1 year	4.8 %
	1 – 3 years	19.9 %
	3 – 5 years	21.6 %
	5 – 10 years	25.0 %
	10 – 15 years	11.1 %
	More than 15 years	17.6 %
Animal ownership	None	24.3 %
	Pet only	66.5 %
	Livestock only	3.5 %
	Both pet and livestock	5.7 %
Internet access	Yes	99.3 %
	No	0.7 %
Smartphone access	Yes	98.1 %
	No	1.9 %
Use of GPS-navigation routinely	Yes	77.3 %
	No	22.7 %
Homeowners insurance status	Not insured	20.0 %
	Under-insured	23.4 %
	Fully insured	56.6 %
Past evacuation experience	None	53.0 %
	Once	18.8 %
	2 – 3 times	21.0 %
	4 – 5 times	5.7 %
	More than 5 times	1.5 %
Past wildfire-induced property damage	Yes	26.7 %
	No	73.3 %
Disaster plan	No plan	46.5 %
	Have a plan but not in a written form	34.4 %
	Have a written plan	19.1 %

Appendix B: Survey data on selected evacuation decisions

Decisions	Classes	Frequency (%)
Evacuation Timing	Evacuate without any official announcement	30.7 %
	Evacuate with a voluntary evacuation order	38.6 %
	Evacuate with a mandatory evacuation order	27.4 %
	Evacuate after smoke is observed	1.2 %
	Evacuate after flames/embers are observed	0.6 %
	Stay and defend	1.5 %
Evacuation Destination	Friend's or family member's house	52 %
	Hotel or motel	28.8 %
	Public shelter	8.9 %
	Secondary residence	4.3 %
	Others	6 %

Appendix C: A comparison table showing the prediction accuracies of regression and machine learning models for evacuee responses

Methods	Evacuation trigger	Evacuation destination	Navigation use
Logistic regression	39.4 %	64.8 %	68.5 %
Decision tree(m_depth, m_samples)	43.8 % (10, 4)	83.0 % (16, 6)	68.2 % (6, 26)
Random forest(n_trees, m_samples)	44.0 % (50, 6)	89.7 % (190, 2)	71.4 % (30, 18)
Support vector machine(function, C, gamma)	42.8 % (Gaussian, 1, 0.01)	90.2 % (Gaussian, 0.1, 1)	71.1 % (Gaussian, 100, 0.01)
Naïve Bayes(variant)	41.3 % (Bernoulli)	60.6 % (Gaussian)	68.4 % (Gaussian)
K-nearest neighbors(n_neighbors)	41.8 % (6)	82.2 % (2)	72.2 % (12)
Neural network(n_nodes, alpha, solver, activation)	43.2 % (12, 0.001, lbfgs, tanh)	78.2 % (16, 0.01, SGD, logistic)	71.8 % (2, 0.0001, lbfgs, logistic)

Data availability

Data will be made available on request.

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